Sanity Checks for Reinforcement Learning Explanations

Abstract. Feature attribution is increasingly proposed as a basis for auditing reinforcement learning agents. We show that an attribution can be stable across training runs, precisely estimated, semantically plausible, and carry no information whatsoever. On a prediction market that is empirically efficient, a PPO agent whose return is statistically indistinguishable from abstention produces attributions with a cross-seed rank correlation of 0.849, unanimous agreement on the most important feature, and a top-5 ranking whose adjacent gaps exceed their own standard errors. Every conventional check passes. We introduce a null-model test, built on the additivity identity that attribution schemes already satisfy, comparing an agent's attribution span against the spans of agents trained where the observation channel carries no information. The test rejects a planted signal at $z = 12.27$ and fails to reject the market agent at $z = +0.23$. We further show that the established sanity check for this purpose, randomising network parameters, does not transfer to reinforcement learning: on CartPole its reference distribution is 42× wider than an environment-level null, and across eight checkpoints on two control tasks it never exceeds $z = 2.45$, including on a converged policy. Finally, the market agent's attribution can be driven from 39.9% to 3.2% of total mass at no measurable cost in return, while the same intervention on a feature that genuinely carries signal destroys 81% of return. Attribution is therefore not identified by behaviour when the attributed feature does not matter.

1 Introduction

A common argument for interpretability is oversight: if we can see which inputs drive a policy, we can decide whether to trust it. This requires that an attribution mean something. It is not obvious that attributions do, and it is less obvious how one would tell.

The difficulty is that attribution methods return an answer regardless. Given any policy and any state, Shapley values [8, 11], integrated gradients [13], and permutation importance all produce a vector of numbers, ranked, signed, and plottable. Nothing in the output distinguishes an attribution reflecting structure the agent exploits from one reflecting nothing at all.

In supervised learning this is handled by randomisation tests. Adebayo et al. [1] compare explanations of a trained model against explanations of a model with randomised parameters, and against a model trained on permuted labels. If the explanation does not change, it does not depend on what the model learned. These tests are standard, and applying them invalidated several saliency methods.

Reinforcement learning has adopted only half of this. Huber et al. [7] apply the parameter-randomisation test to saliency maps for Atari agents, and observe that the data-randomisation test has no obvious RL analogue: there is no labelled dataset to permute, and it is unclear how a sequential decision maker should respond to artificially corrupted features.

This paper supplies that analogue, and shows that the half already adopted does not work.

Contributions.

1. **An environment-level null for RL.** Rather than corrupting the model or the labels, we corrupt the environment's *observation channel*, leaving dynamics and reward intact. The agent trains normally, on a real objective, with nothing to condition on. This requires no access to task internals (Section 3).
2. **The established null fails in RL, measurably.** Parameter randomisation yields a reference distribution 42× wider on CartPole and degenerate on Acrobot. Across eight checkpoints it never exceeds $z = 2.45$, including on a converged CartPole policy scoring 404 of 500, while the environment null reaches $z = 117$. The two disagree on five of eight (Section 5.5).
3. **A validation protocol with ground truth.** Explanation evaluation is hampered by the absence of ground truth [5]. We construct environments where the correct attribution is known by construction (Sections 4.1 and 5.2).
4. **A worked case** of an uninformative explanation passing every conventional check (Section 5.1).
5. **Attribution is not identified by behaviour.** An auxiliary training penalty drives the market agent's dominant feature from 39.9% to 3.2% of attribution mass at no measurable cost in return; the same intervention on a load-bearing feature costs 81% of return (Section 5.6).

What we do not claim. The argument that RL attribution should target outcomes rather than policy outputs is due to Beechey et al. [3], whose SVERL framework unifies behaviour, outcomes and prediction, and whose FastSVERL [2] addresses the scalability, off-policy and temporal problems we only name. Our test adapts the data-randomisation test of Adebayo et al. [1]. Deletion-based fidelity evaluation is established practice in XRL [5, 9]. Our contribution is the RL construction, the demonstration that the incumbent alternative fails, and the empirical consequences.

2 Related work

Shapley values in RL. Beechey et al. [3] give a theoretical account of Shapley-value explanation for RL, showing that earlier applications explained the wrong object and proposing characteristic functions built on agent performance. Their framework identifies three explanatory targets: behaviour, outcomes, and prediction. FastSVERL [2] makes the approach tractable and addresses temporal dependence and off-policy data. We adopt the outcomes framing and implement it independently; we do not extend the theory.

Sanity checks. Adebayo et al. [1] introduced parameter- and data-randomisation tests; Binder et al. [4] examine their limitations. Huber et al. [7] apply the parameter test to perturbation-based saliency for Atari agents and note the data test lacks an RL analogue. We propose one, and show the parameter test is the weaker of the two in this setting.

Manipulating explanations. Slack et al. [12] construct adversarial classifiers that detect the off-manifold points LIME and SHAP query and behave differently on them, yielding innocuous explanations for biased models. That is an attack on the explainer's sampling. Our steering result differs in kind: we train a normal agent with an invariance penalty, the resulting explanation is *correct* about the resulting model, and the point is that behaviour does not pin the explanation down. It is a non-identifiability result rather than an attack.

Uncertainty in Shapley estimation. Goldwasser and Hooker [6] certify top-k Shapley rankings against Monte Carlo error. These guarantees concern the estimator. Section 5.4 shows they are orthogonal to whether the explained model learned anything.

Evaluation in XRL. Surveys note that the absence of ground-truth explanations is a central obstacle, and that perturbation-based fidelity is the common recourse [5, 9]. We use deletion curves in that established sense and do not claim them as a contribution.

3 Method

3.1 The statistic

Let $v(S)$ be the expected episode return of an agent observing only the features in $S \subseteq N$, with features outside S replaced at every timestep by draws from a reference distribution. Define the *attribution span*

$$\Delta \;=\; v(N) - v(\emptyset). \tag{1}$$

Shapley values [11] and integrated gradients [13] satisfy additivity identities (efficiency and completeness respectively) under which per-feature attributions sum to Δ. The span can therefore be read off either.

It is worth stating precisely what Δ is. It is *not* a Shapley quantity: it is the difference between two masked rollouts, and depends only on the masking scheme. Efficiency tells us the Shapley values happen to sum to it. This has a practical consequence we exploit: Δ costs two evaluations rather than $2^{|N|}$.

3.2 The null

The test compares an observed Δ against the distribution of Δ for agents that had nothing to learn. We construct these by corrupting the observation channel.

Blind-observation null. *Replace every observation with an independent draw from a fixed reference distribution matched to the environment's observation moments. Leave dynamics and reward untouched. Train normally.*

Such an agent faces the real task and receives real return; it simply cannot condition on anything. This is the RL counterpart of training on permuted labels, and unlike parameter randomisation it yields a *normally trained* policy: the failure mode that occurs in deployment rather than an artificial one.

Matching the reference moments matters. A network fed out-of-range inputs fails for reasons of scale rather than information, which would be a different experiment.

3.3 The test

Given Δ_{obs} and null draws $\Delta_1, \ldots, \Delta_n$ with mean $\bar{\Delta}$ and standard deviation s, we report

$$p_{\text{rank}} = \frac{\#\{i : |\Delta_i - \bar{\Delta}| \geq |\Delta_{\text{obs}} - \bar{\Delta}|\} + 1}{n+1}, \quad z = \frac{\Delta_{\text{obs}} - \bar{\Delta}}{s}, \tag{2}$$

and require both to reject.

Two statistics are reported because neither suffices. The rank statistic assumes no distributional shape but bottoms out at $1/(n+1)$: with $n = 8$ it cannot reject at $\alpha = 0.05$ however extreme the observation, and in an early run it reported $p = 0.111$ for a signal lying 10.5 standard deviations outside the null. The normal statistic has resolution but assumes a shape the null need not have. Requiring both keeps the test conservative in the direction that matters.

Degenerate nulls. If every null draw is identical, $s = 0$ and z is undefined. This is not an absence of information but its opposite: any deviation is maximally surprising. We return $z = \pm\infty$ and surface the degeneracy explicitly. It occurs in practice (Sections 5.5 and 5.7) whenever a task lets a blind agent converge on a single behaviour.

4 Experimental setup

4.1 Environments with known answers

Explanation evaluation lacks ground truth [5]. We construct three synthetic corpora in which the correct answer is fixed by construction. Episodes are fixed-length with no early termination, so the horizon is controlled exactly.

- **Planted signal.** One of 18 features is informative about the binary outcome; the rest are noise or constants.
- **Null.** Identical shapes and frictions, no informative feature. The optimal policy is to abstain.
- **Decoy.** One feature correlates 0.989 with the outcome in the first 60% of episodes and 0.006 in the remainder.

The claim about the planted corpus needs care. The planted feature is the only one informative about the *outcome*. Others can still affect the *return* without predicting anything, because return depends on behaviour: an agent that cannot tell where it is in an episode acts incoherently and pays transaction costs. The test is that the planted feature ranks first with the majority of attribution mass, not that everything else is zero.

4.2 A market with terminal ground truth

Our main empirical setting is a binary contract settling to \$1.00 if BTC is higher after 15 minutes and \$0.00 otherwise, on a regulated prediction market. We use 6,425 settled contracts over 68 days (89,950 transitions, outcome rate 0.4999), with a walk-forward split by time and purge gaps; the test split is evaluated once.

This instrument was chosen for one property: every episode resolves to a known value, so a critic's $V(s)$ can be checked against realised settlement frequency. Agents are trained with PPO [10]. State is 18 causal features. Actions are target inventory in $\{-1, 0, +1\} \times q_{\max}$. Reward is the change in mark-to-market equity net of costs, which telescopes exactly to net episode P&L.

Transaction costs use the exchange's published taker schedule, $\lceil 0.07\, C\, P(1-P) \rceil$, maximised at $P = 0.5$ where this contract trades by construction. With the measured one-cent modal spread this is a 9% round-trip hurdle on a 50-cent notional, and it determines most of what follows.

4.3 Control tasks

CartPole-v1 (4 features) and Acrobot-v1 (6 features) provide settings with no relationship to trading and well-understood optimal policies. Both have observation spaces small enough to enumerate all $2^{|N|}$ coalitions, so per-feature Shapley values there are exact.

Table 1: Test split, 1,284 held-out episodes. p-values are paired bootstrap against always-flat on the same episodes, necessary because per-episode P&L has a standard deviation near 50 against differences near 1.

Policy	Mean P&L	Trades/ep	p vs flat
always-flat	+0.000	0.00	—
buy-and-hold	−1.608	1.00	0.23
PPO (5 seeds)	$\mathbf{-0.595 \pm 0.176}$	1.37	**0.10**
logistic (refit)	−5.729	2.52	<0.0001
mean-reversion	−11.615	4.94	<0.0001
random	−18.532	9.33	<0.0001

4.4 Reproducibility

All results derive from public endpoints with no API keys. Hyperparameters were fixed a priori from synthetic sweeps where the correct answer is known analytically, not tuned on validation performance. The full pipeline is one command; 226 tests cover the environment, the attribution implementations, and six no-lookahead properties verified by mutation testing. Every numerical claim in this paper is asserted against its source artifact by a script that runs as the pipeline's final stage.

The exchange serves a moving window, so re-running the ingest returns a different set of contracts. A cold rebuild five days after the original produced 6,425 episodes against 6,428, sharing 5,960 tickers; *overlapping episodes are byte-identical on every field.* Ingest is deterministic per contract and what changes is membership. All conclusions reproduced (Section 5.8).

5 Results

5.1 An explanation that passes every check and says nothing

The market is empirically efficient: its implied probability has a weighted mean absolute calibration error of 0.0172 across ten bins, and spot return correlates 0.476 with the outcome but only 0.016 with the residual after the price is accounted for. No policy beats abstention (Table 1); losses track turnover.

PPO's shortfall is its own. On synthetic data containing no signal by construction the same agent scores -0.47 ± 0.47; here it scores -0.595 ± 0.176. Nothing is left to attribute to the market.

Yet its explanations look entirely reasonable. Across five seeds whose pairwise performance differences are statistically indistinguishable in all ten pairs: behaviour attributions have cross-seed rank correlation 0.849 (min 0.761) with 100% agreement on the top feature and 100% sign agreement; the top-5 ranking is fully certified, all four adjacent gaps exceeding their combined standard errors; and the ranking is semantically legible, time-to-expiry dominating, which reads as an agent that learned when not to trade.

Table 2: Null-model test on the market. Null from 24 blind-trained agents, span $+8.19 \pm 5.11$.

Case	Span	z	Verdict
planted signal	+70.89	**+12.27**	informative
null corpus	+11.86	+0.72	not distinguishable
real market	**+9.37**	**+0.23**	**not distinguishable**

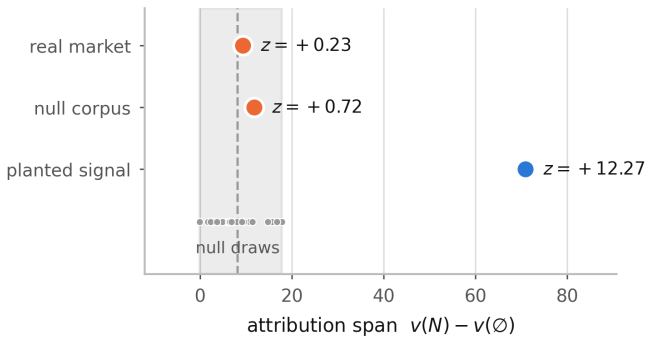

Figure 1: Attribution spans against the null distribution. The shaded band is the range over 24 agents trained where the observation channel carries no information. The market agent sits at its centre.

The null test disagrees (Table 2, Figure 1).

The test has power and specificity. The market agent's explanation sits at the centre of the distribution of explanations of nothing.

We note one consequence for our own reporting. An earlier draft stated that "the whole observation is worth +5.914 per episode against being blind" and presented it as a finding. Blind agents produce $+8.19 \pm 5.11$. It was noise.

5.2 Validation against ground truth

On the planted corpus, outcome attribution ranks the planted feature first with 55% of total attribution mass. This check is unavailable on real data and is what licenses the rest.

On the decoy corpus, an agent trained where the decoy predicts the outcome (in-sample +47.24, held out −6.83) is explained on held-out episodes where the decoy is provably worthless (Figure 2). Per-decision attribution of $\pi(a \mid s)$ gives it 6.5% of mass, rank 3 of 18; outcome-based attribution gives it 1.1%, rank 10.

Both statements are true. The policy does key on the decoy, so per-decision attribution is correct about behaviour and misleading about value. An auditor asking what an agent relies on that does not work gets the wrong answer from it.

Deletion curves agree: removing features in the order each ranking considers important, the outcome ranking degrades return faster (AUC −187.8 against −165.3; a random ranking gives +366.6).

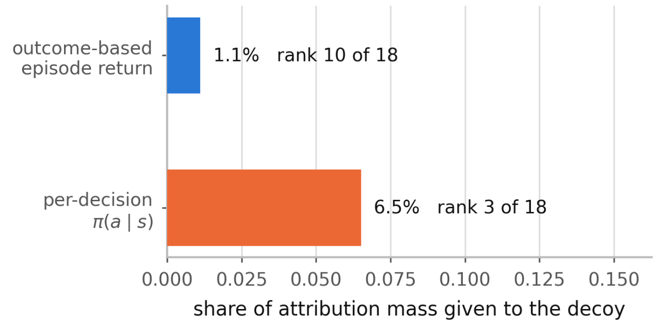

Figure 2: A feature that drives the policy but earns nothing, on held-out episodes where it is provably uninformative.

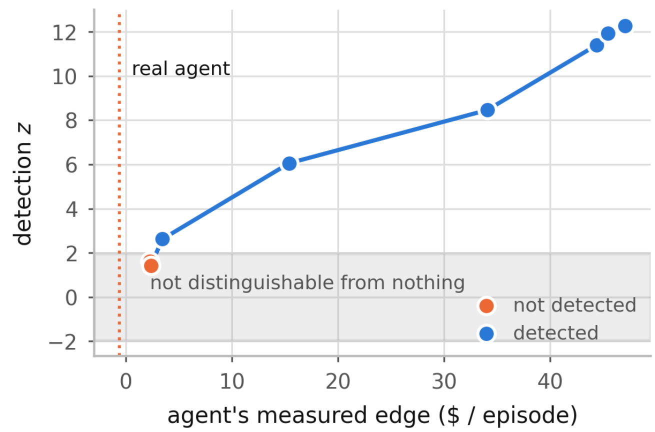

Figure 3: Detection z against the agent's measured edge. The shaded band is where an explanation is not distinguishable from an explanation of nothing.

5.3 How much edge is required

Sweeping planted-signal strength calibrates the test (Figure 3). Detection begins near +3.45 per episode of *measured* edge. The market agent earns −0.595: an order of magnitude below threshold and of the wrong sign. These edges are in-sample and therefore optimistic; the threshold is a lower bound.

5.4 Estimation guarantees do not substitute

Goldwasser and Hooker [6] certify that a top-k ranking is stable under Monte Carlo noise: a guarantee about the estimator. The market agent's top-5, with standard errors near 0.001 on a leading value of 0.26, has all four adjacent pairs separated beyond their combined error and is fully certified by that criterion. The same explanation fails the null test at $z = +0.23$.

A ranking can be certified stable while the explanation it ranks is indistinguishable from an explanation of nothing. Only one of these two properties is commonly reported.

Table 3: Null construction on two control tasks.

Environment	Parameter rand.	Environment rand.	Ratio
CartPole-v1	$+55.34 \pm \mathbf{138.92}$	$-1.35 \pm \mathbf{3.29}$	42×
Acrobot-v1	$+64.03 \pm \mathbf{124.94}$	$+0.00 \pm \mathbf{0.00}$	degenerate

Table 4: Steering one feature's attribution.

Corpus	Attribution	Return	p
market	39.9% → **3.2%**	−3.49 → −1.65	**0.66**
planted signal	46.5% → **1.5%**	+45.04 → **+7.92**	<0.001

5.5 The established null does not transfer to RL

We construct both nulls for the same task (Table 3, Figure 4). Shapley values here are exact.

A randomly initialised policy behaves arbitrarily, so masking its inputs moves return unpredictably and the reference distribution is very wide. The resulting test has almost no power. Across eight checkpoints the parameter null never exceeds $z = 2.45$, including on a converged CartPole policy scoring 404 of 500, while the environment null reaches $z = +117.6$. The two disagree on five of eight. The choice of null is not a detail; it decides the verdict, and the incumbent construction is the one that fails.

Acrobot additionally shows the test tracking *whether there is anything to explain* rather than agent quality. Its agent fails completely until it learns (−500 at 10%, 25%, 50% of training), the span is exactly 0.00 there, and the test correctly declines. CartPole behaves differently because its observations matter at every skill level. We do not claim undertrained agents generally produce empty explanations; they do not.

5.6 Attribution is not identified by behaviour

We add an auxiliary penalty during training on the divergence between $\pi(\cdot \mid s)$ and $\pi(\cdot \mid s')$, where s' has one feature resampled from the batch marginal: the same interventional perturbation the attribution measures (Table 4, Figure 5).

The attribution is equally suppressible in both cases (92% and 97%). What differs is the price: where the feature is the planted signal, suppressing it destroys 81% of return; on the market it costs nothing measurable. Explanations are steerable at no cost exactly where they are not tracking anything, corroborating Section 5.1 by a route sharing no machinery with the null test. The control matters: had steering succeeded on both corpora, the penalty would merely be defeating the attribution method.

5.7 What the verdict does and does not depend on

Attribution family. Integrated gradients [13], sharing no machinery with Shapley, ranks features at 0.981 correlation with Shapley across five seeds and is comparably seed-stable (0.750 against 0.800). Per-feature attributions are not a Shapley artefact. IG cannot test the outcome-level claim: episode return is not differentiable through the environment, so IG has no outcome analogue. Its behaviour-level span reports the market agent as informative ($z = +4.58$), which is consistent rather than contradictory and is precisely the behaviour-versus-outcome distinction of Beechey et al. [3]. The agent's behaviour does respond to its observations; that responsiveness earns nothing.

Credit assignment. Leave-one-out ($v(N) - v(N \setminus i)$) and only-one-in ($v(\{i\}) - v(\emptyset)$) are the extremes of the average Shapley takes, and neither satisfies efficiency, so each carries its own total. On the planted signal all three fire ($z = +10.66$, $+10.96$, $+2.29$); on the market none does ($z = -0.42$, -1.57, -0.72). The verdict is scheme-independent. *The per-feature ranking is not*: leave-one-out and only-one-in rank features at only 0.309 correlation with each other on the same agent. Whether there is anything to explain is robust; which features matter is not, and a ranking from a single scheme should not be reported as the answer.

Horizon. We hypothesised that the parameter null's excess variance arises because randomising weights perturbs the measurement's *domain*, return being a functional of the policy-induced state distribution, and that this compounds along the trajectory. The prediction is that its variance grows with horizon faster than the environment null's. It does not: over horizons 4 to 28 both grow near-linearly at indistinguishable rates ($\alpha = 0.94$, $r^2 = 0.995$ against $\alpha = 0.83$, $r^2 = 0.998$), the ratio flat at 1.7–1.8×. The hypothesis is rejected and the 42× gap remains unexplained.

The experiment did establish something else. At horizon 56 the environment null collapses to a standard deviation of 0.04 with two distinct values across sixteen agents: every blind agent converged on the same abstaining policy. This is the Acrobot degeneracy again, and it now has a mechanism: the environment null degenerates when the task affords enough time to learn that inaction is optimal.

5.8 Reproduction from scratch

All figures above come from a corpus rebuilt after deleting every cached file. An earlier corpus, collected five days before and sharing 93% of its contracts, gave PPO -0.418 ± 0.260 ($p = 0.129$) against -0.595 ± 0.176 ($p = 0.100$), and null-test z of $+12.27/+0.72/-0.15$ against $+12.27/+0.72/+0.23$. Every conclusion is

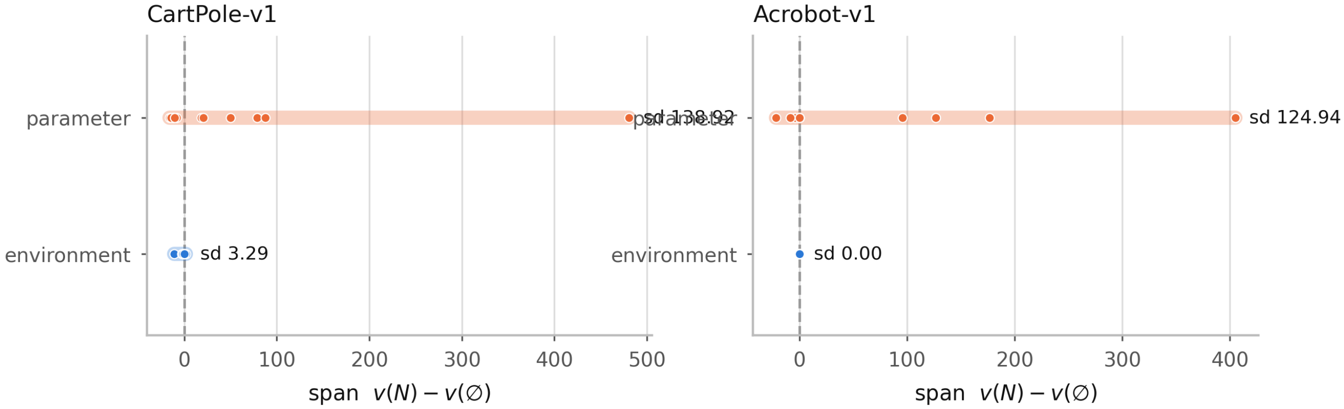

Figure 4: Spread of each null construction, on both control tasks. Width determines power: a very wide reference distribution cannot reject anything.

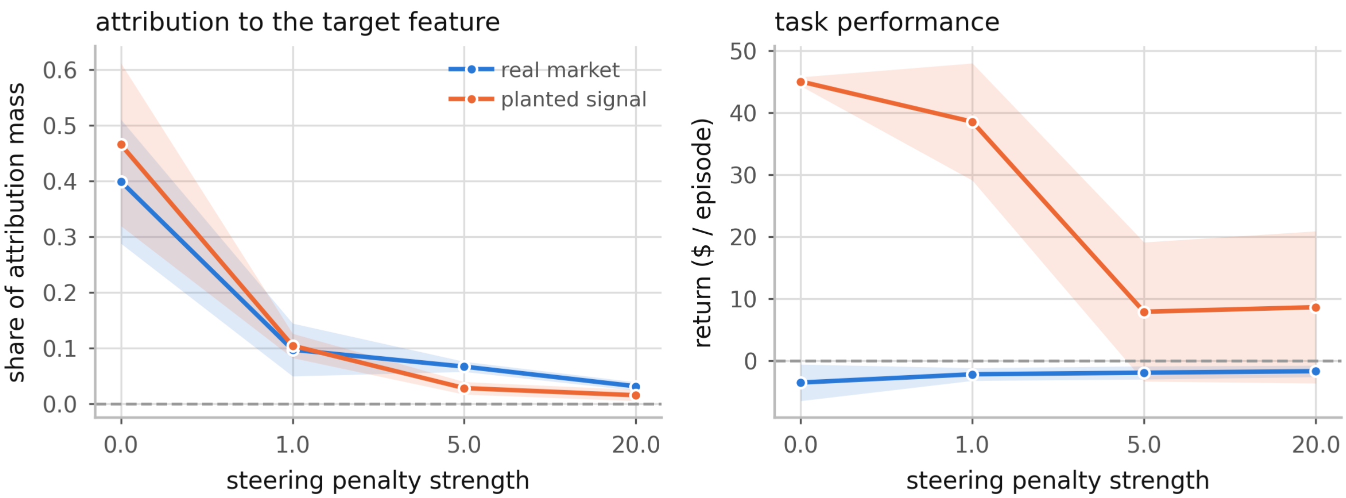

Figure 5: Attribution is equally suppressible on both corpora (left). Only where the feature is load-bearing does suppressing it cost return (right).

identical on both. Buy-and-hold moved from $+0.068$ to -1.608, confirming its earlier positive figure reflected that period's 52.3% outcome rate rather than an edge.

6 Limitations

The construction is adapted. The test is the data-randomisation test of Adebayo et al. [1] with an RL-appropriate null; the outcomes framing is Beechey et al. [3]'s; deletion-based fidelity is standard.

The span measures information in aggregate. An explanation could clear the test and still misattribute among features; Section 5.7 shows the ranking is scheme-dependent even when the verdict is not.

Temporal credit assignment is named, not solved. Masking a feature for a whole episode measures whether it mattered, not when. This is FastSVERL's [2] territory.

Off-policy attribution is not handled. Masked coalitions induce a different policy and therefore a different state distribution; estimates are on-policy for the masked policy, not counterfactuals about the unmasked one.

The 42× gap is unexplained. Our proposed mechanism was tested and rejected. A remaining conjecture, consistent with all three environments but untested, is that the parameter null's width depends on how often a randomly initialised network is an accidentally competent policy: sometimes on CartPole and Acrobot, never on a market with no exploitable signal.

Scope. One real domain and two classic-control tasks; one attribution family for the outcome-level tests. Corpus membership is not reproducible across time, though content is.

7 Conclusion

An attribution can be stable, precisely estimated, plausible, and empty. The checks practitioners apply — consistency across runs, separation beyond estimation error, semantic legibility — do not detect this, because none of them ask whether the explained agent learned anything. A null-model test does, using a statistic the attribution framework already provides. The established test for this purpose does not survive the move to reinforcement learning, and we recommend against its use there without the measurement in Section 5.5.

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